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Empanada Documentation

A GUIDE ON USING THE NAPARI PLUGIN EMPANADA FOR ORGANELLE SEGMENTATION

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Introduction

Aim

This document is an illustration on the use of the in-built models of the Napari plugin *Empanada* for organelle segmentation using FIB-SEM images. The task was undertaken while working in the AI directed Voxel Extraction (AIVE) Project within the WEHI Research Computing Platform.

Data Source

The test images used are FIB-SEM images of cells in the form of multi-stack tiff files that have been provided by the WEHI Research Computing Platform.

Physical System Specs

Windows 11

Processor: 11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz (3.00 GHz)

RAM: 16.0 GB

Downloading Empanada

To use the Empanada plugin, Napari must be downloaded. But keep in mind that Empanada-Napari only supports Python 3.9 and uses Napari version 0.4.8 while the current version is 0.6.4. For this reason, it is advised to use a virtual environment to use this plugin.

For this documentation a conda environment which used Python 3.9 was implemented.

The following link provides step by step instructions on how to download empanada-napari.

[Installation — empanada-napari-v1.2 1.2 documentation]

Note: Installing the pluggin from the napari GUI might not work so use the command line, this will install and uninstall dependencies appropriately.

Once Napari is downloaded and opened it should look like the image on the right (Figure 1).

If any WARNINGS are given during the launch then simply close it and start again.

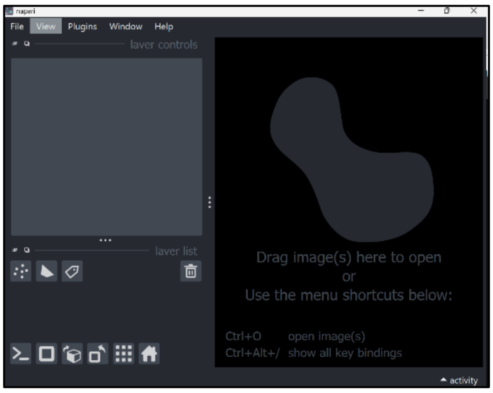

Figure 1: Napari Viewer

The plugin can be located through the top left options.

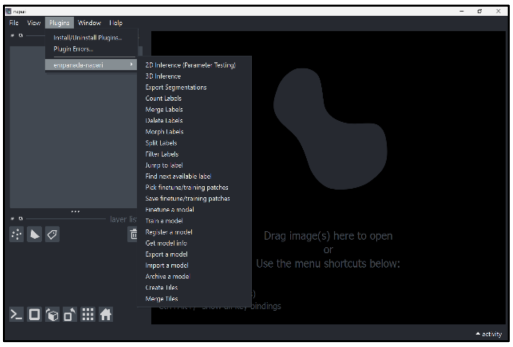

Figure 2: Locating Empanada in Napari

To load an image into napari click CTRL+O and select the file. You can open multiple images and click through them with the layers list on the bottom left.

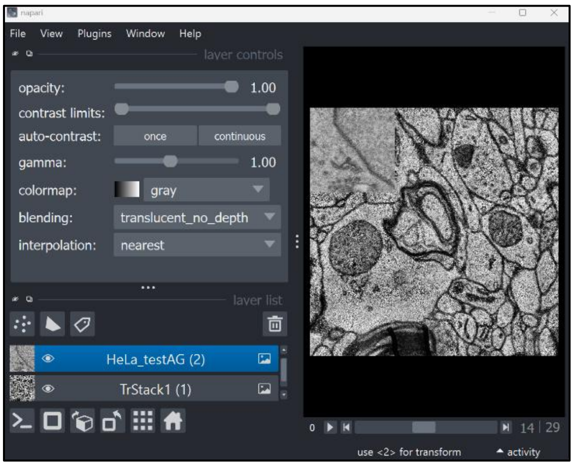

Figure 3: Opening Images and Navigating layers

Running Empanada

To perform segmentation through Empanada you can use the built in models or import your own models onto napari.

For this documentation we will be using the models that are already provided.

Empanada provides 4 models:

1. DropNet_base_v1 (for lipids)
2. MitoNet_v1 (for mito)
3. MitoNet_v1_mini (for mito)
4. NucleoNet_base_v1 (for nuclei)

To get specific information on the model choose the option “Get Model Info”. It will print out model specifics into the terminal.

Organelle Segmentation

There are two ways you can perform segmentation of organelles in Empanada- 2D inference and 3D inference.

For simplicity the MitoNet_v1 and NucleoNet_base_v1 models - which have been used for segmentation - will be referred to as MitoNet and NucleoNet respectively.

2D Inference (Parameter Testing)

2D inference runs a model of the user’s choice on a single slice/frame of the image.

To run the model simply navigate to 2D Inference (Parameter Testing) and select the image layer and model of your choice. The user can then go through the parameter options provided and enter their choices.

The slice that the model will run on is the slice that is currently on the screen. A sliding bar on the screen will indicate the slice (for example 12/30 mean slice 12).

Figure 4: 2D Inference Viewer

Please note the models run on the images were done using the default parameters provided.

Once the user runs the 2D inference the segmenting process will begin and as an output the segmentation for the slice chosen will be provided as a new layer in the layers list.

The image below shows slice 15 before and after the NucleoNet model has been run on it:

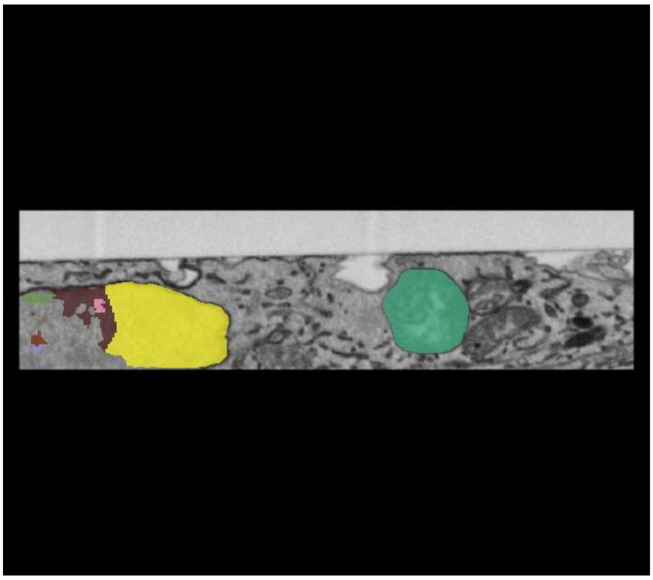

Figure 6: Nucleus1 Image, Slice 15, 2D Inference

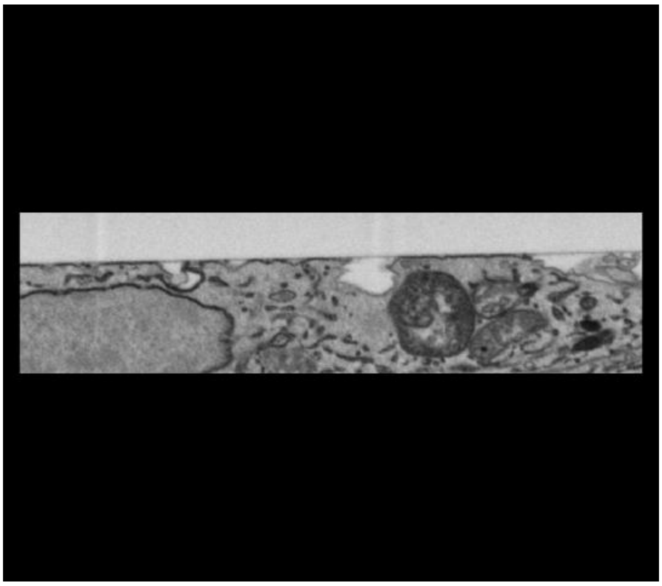

Figure 5: Nucleus1 Image, Slice 15

The output layer is in the form of a mask where only the segmented area is coloured in and the rest of the image is dark. By adding the original image's layer to it you can view the whole segmented image.

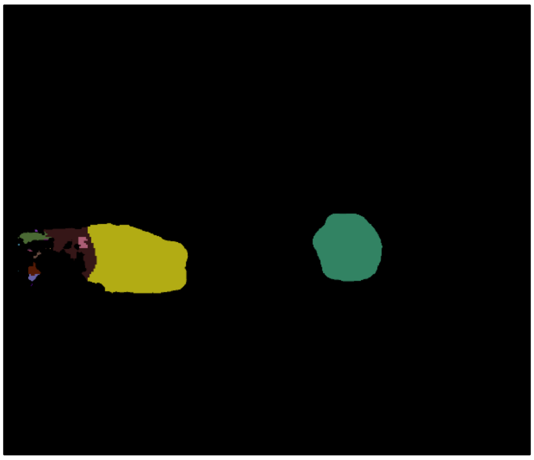

Figure 7: Nuclues1 Slice 15, 2D Inference Segmentation Layer

Batch Mode

To run the model on all the slices in the image individually select the 'Batch Mode' option before running the model. This will provide a segmented layer for the entire FIB-SEM image.

As an output a new layer in the layers list will have all the segmentations for each slide for the user to view through.

3D Inference

3D inference allows the user to run the model on the entire image at once. Unlike 2D inference the model is not applied independently to each slice.

While setting up the model to run is similar to 2D inference, there are more options available through 3D inference such as Stack parameters, Ortho-plane parameters and Zarr storage. Some of these new options are optional.

The Zarr storage option allows the user to save the segmentation as Zarr batches in the directory of choice. This allows the user to run the models faster as without the Zarr storage the memory required to run the model increases.

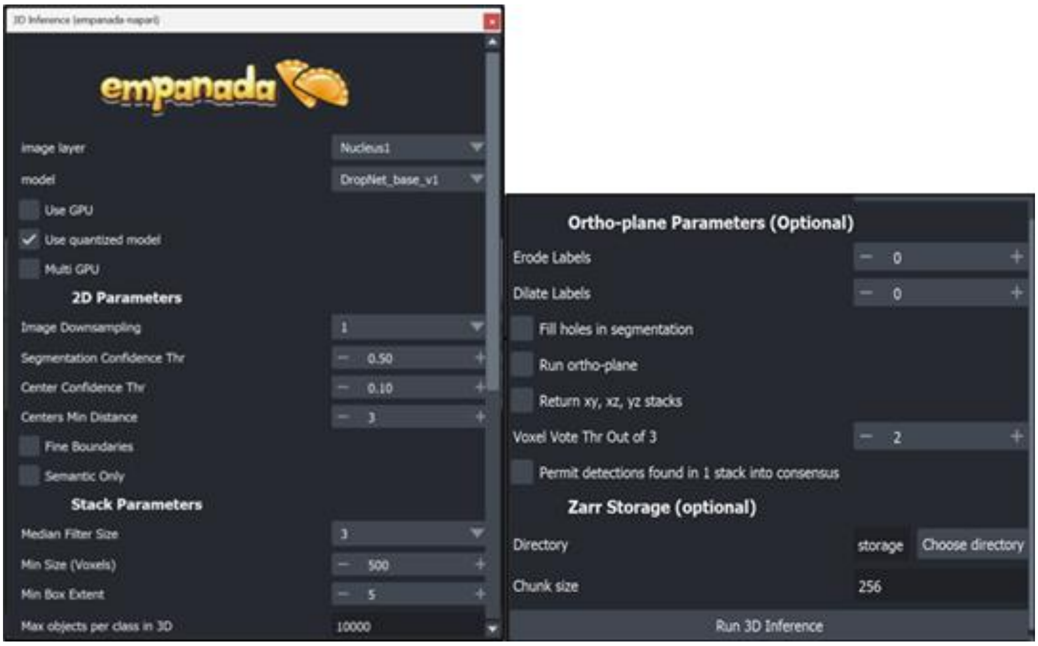

Figure 8: Napari 3D Inference Interface

Once the model has been run, just like the 2D inference a new layer will appear with the segmentation results.

A fact to note is that in some cases when running 3D inference the segmentations in the final slices are not recorded by the model. The following example is by running the NucleoNet Model on the HeLa_testAG image.

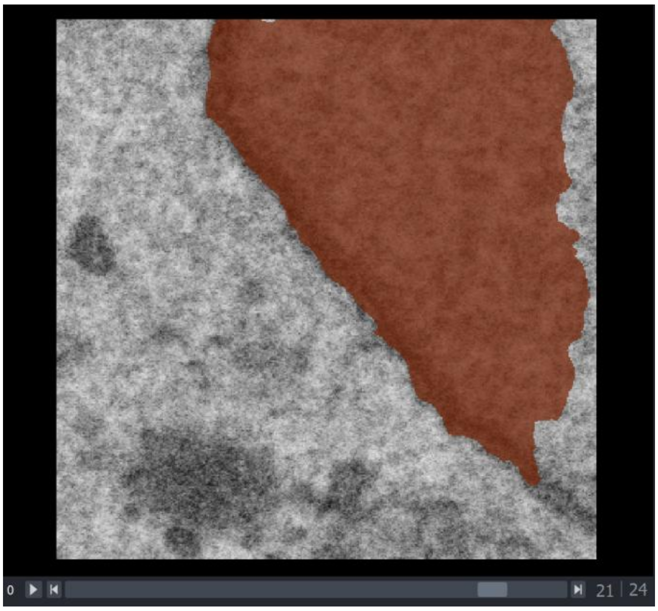

Figure 9: Slice 21, 3D Inference on Hela-testAG with NucleoNet Model

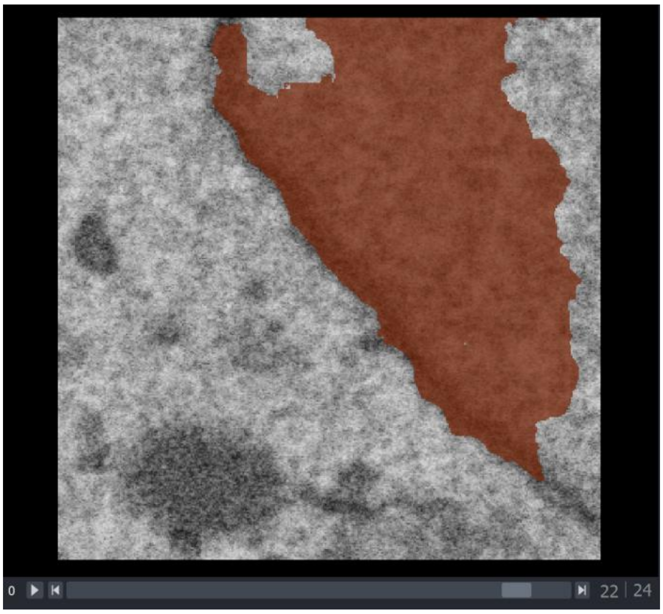

Figure 10: Slice 22, 3D Inference on Hela-testAG with NucleoNet Model

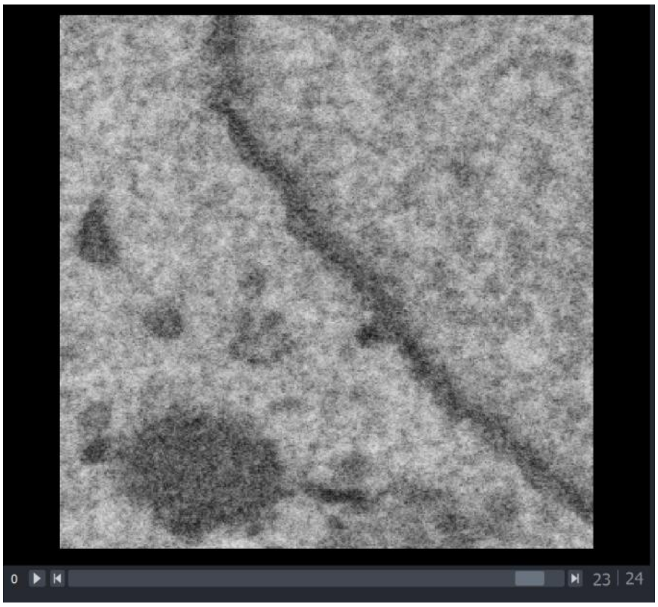

Figure 11: Slice 23, 3D Inference on Hela-testAG with NucleoNet Model

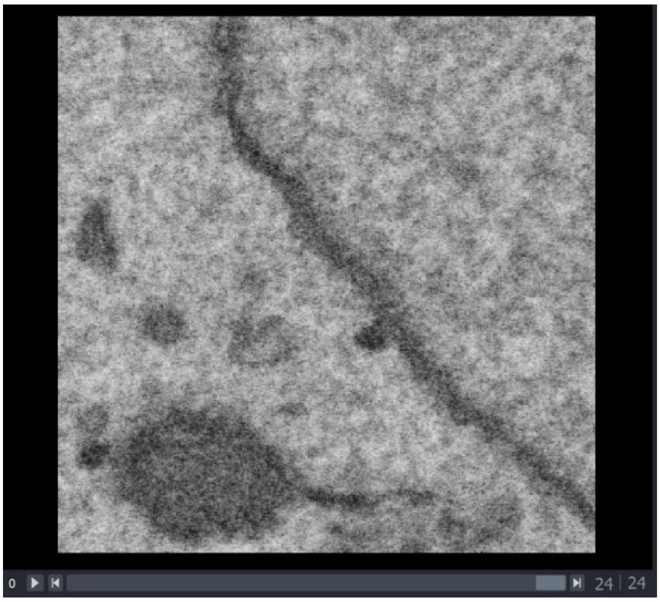

Figure 12: Slice 24, 3D inference on Hela-testAG with NucleoNet Model

In the results the model's predictions for the nuclei which are present in slides 22 and prior vanish later.

Comparing Results

To compare results from the three techniques Frame 15 from the Nucleus1 image after running the NucleoNet model will be used:

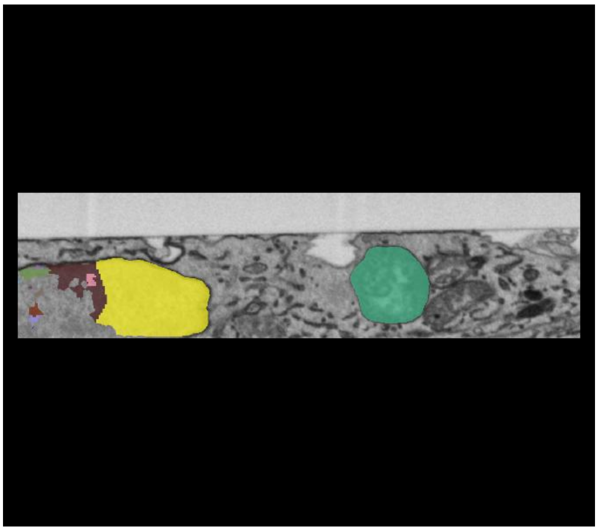

Figure 13: Frame 15, Nucleus1, 2D Inference with no Batch Mode

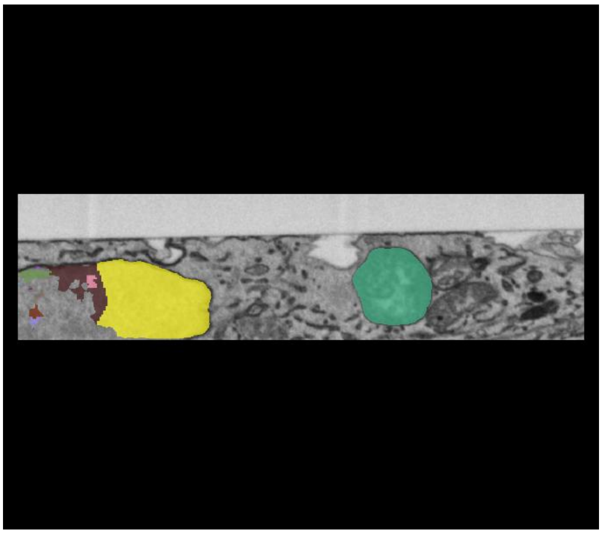

Figure 14: Frame 15, Nucleus1, 2D Inference Batch Mode

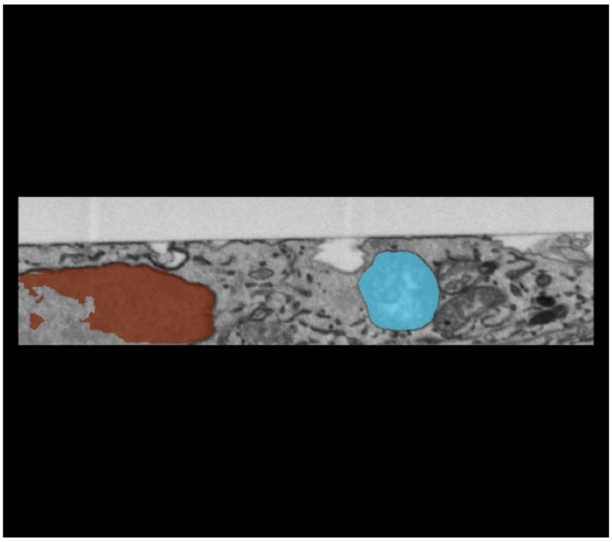

Figure 15: Frame 15, Nucleus1, 3D inference

From the images above it can be inferred that there is no difference between the results from the 2D inference with and without batch mode. This is as expected as the model is applied independently to each frame.

The difference that can be observed is between the 2D inference and 3D inference where each colour represents an object and the 2D inference appears to have far more coloured in structures marked in the same area where the 3D inference only has one.

While this instance shows a similar output for both inference modes. The images below do not share the same result.

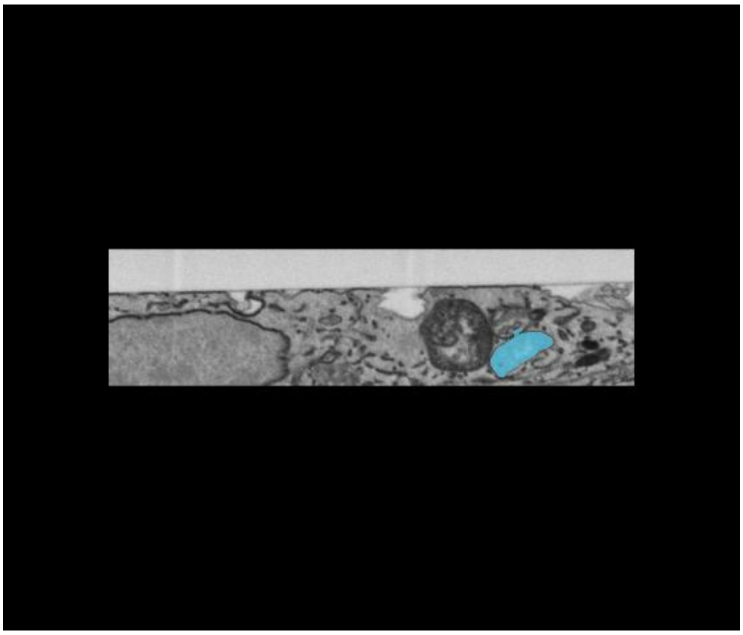

Figure 16: Slice 15, 3D Inference on Nucleus 1 - MitoNet Model

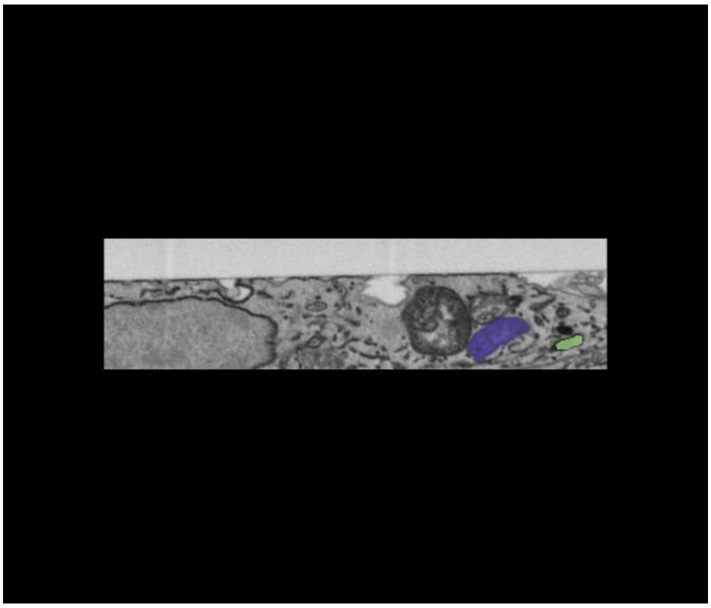

Figure 17: Slice 15, 2D Inference on Nucleus1- MitoNet Model

Both images are from running the MitoNet Model on Slice 15 of the Nucleus1 image and we can see that the 2D inference produces 2 mitochondria on the same slice that 3D inference produces 1.

Changing the Data Type

As per the requirements of the user the data type for the segmentations can be changed. The default option is 32-bit.

The data type can be changed by selecting the layer and navigating to convert data type.

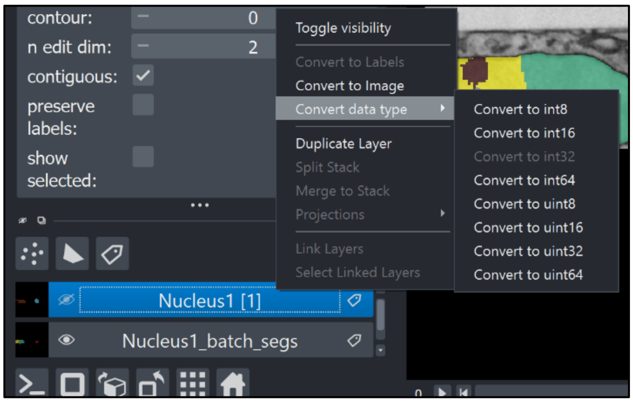

Figure 18: Changing Data Types

Exporting the Segmentations

Empanada allows the segmentations to be saved as masks in .tif file formats. To save an image navigate to the Export Segmentations option of the plugin. Here you select the image layer (the layer with the original images), and label layer (the layer with the segmentations), enter a folder name you want the segmentations to be saved as and choose the directory for the folder.

The user also has the option between 2D images and 3D images. The first saves the segmentation of each slice separately while the second saves a multi-stack .tif file.

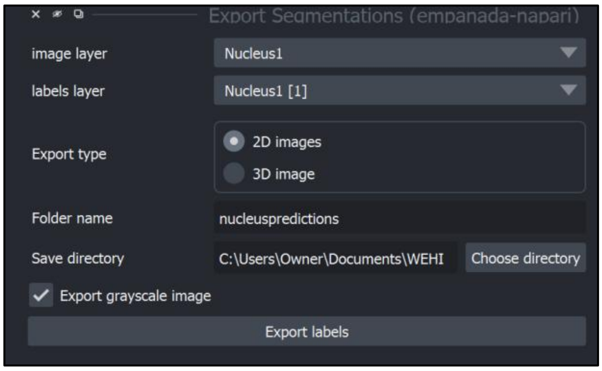

Figure 19: Export Segmentation View in Napari

Technical Factors to Note

When using Empanada-Napari the following technical factors/issues are to be considered:

1. The first time a model is used it is downloaded which will take a few minutes. Once the model has been downloaded once, subsequent uses will be faster
2. A technical issue that came up is if a model failed to download the first time it was not possible to use it at all later. For this reason, the DropNet_base_v1 model was unable to be used in segmentation.
3. As running the model can require extensive memory avoid having other programs running in the background when applying the model.

Resources

For further information regarding Empanada and on how to use it the following website can be used:

https://empanada.readthedocs.io/en/latest/